

Vijayaragavan S

MACHINE LEARNING / AI ENGINEER

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Computer Science undergraduate graduating in 2027, working on computer vision, time-series models and deployment on constrained devices. First author of an IEEE-published pothole-detection paper. Recent work includes a 925-subject anaemia screening study and an offline Android inference app.

TECHNICAL SKILLS

Languages and libraries: Python, SQL, Java, C; PyTorch, TensorFlow, Scikit-learn, OpenCV

Machine learning: CNNs, U-Net, temporal convolution, attention, transfer learning, ensembles

Evaluation: Stratified cross-validation, ROC-AUC, calibration, bootstrap intervals, subgroup analysis

Data and deployment: Pandas, NumPy, Streamlit, Flask, Docker, LiteRT / TensorFlow Lite, Git

PROJECTS

Anaemia Screening from Smartphone Eye Images

2026 | PyTorch, OpenCV, Scikit-learn, LiteRT, Java

- Compared four CNN/transformer architectures on 925 subjects using five-fold cross-validation, with calibration and decision thresholds fitted only on validation data.
- Reached **0.881** pooled out-of-fold ROC-AUC (95% CI: 0.860-0.903) with a three-seed MobileNetV2 ensemble and test-time augmentation.
- Evaluated cross-dataset transfer, population subgroups and low-resolution inputs; identified weak transfer between cohorts and used resolution checks in the research app.
- Compressed a classifier from **8.57 MB to 2.54 MB** while retaining 0.866 AUC in an out-of-fold precision study; built an offline Android app with U-Net segmentation.

Real-Time Pothole Detection | Published Research

2025-Present | Python, TensorFlow Lite, C, ESP32 | [Source code](#)

- Built a dual-accelerometer pipeline with 3,245 labeled signal windows; a temporal convolutional network with attention achieved **96.7%** test accuracy against six baselines.
- Used knowledge distillation, pruning and quantization to retain **95.8%** accuracy in a **342 KB** edge model with **47 ms** inference on ESP32.

Carbon Emissions Intelligence Platform

2025-Present | Python, Pandas, Scikit-learn, Flask, PostgreSQL | [Source code](#)

- Built emissions-data ingestion, cleaning and validation workflows; developed trend forecasting and interactive Plotly dashboards for reporting.
- Exposed analytics through Flask REST APIs and containerized the application with Docker.

PUBLICATION

Vijayaragavan Sivasamy et al. (first author). Real-Time Pothole Detection Using Dual Accelerometer Sensors with Temporal Convolutional Networks and Attention Mechanisms. IEEE conference publication, 2026.

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EDUCATION

B.Tech, Computer Science and Engineering | 2023-2027 (Expected)

SRM Institute of Science and Technology, Chennai | **CGPA: 9.46/10**

Relevant coursework: Machine Learning, Data Structures and Algorithms, DBMS

Certifications: Java - NPTEL